CHAPTER

Design Requirements

2

2.1 INTRODUCTION

As discussed in Chapter 1, products can be broken down into a set of product elements (i.e., components, joints, and in-process structures). Each of these product elements contributes to a function or feature of the product. The performance of each product element influences the performance of the product. Furthermore, the reliability of the product depends on the reliability of the product elements. If a product element fails, then some function or feature will degrade or fail. Finally, the cost to produce a product depends on, among other things, the costs to make or purchase the product elements.

The performance, reliability, and cost of a product element depend on its physical construction and the properties of its constituent materials. The physical construction of a product element refers to its shape and dimensions. It also includes the manner in which multiple materials are incorporated into a product element. For example, some components are composed of a base material with one or more coatings applied over the surface, as shown in Figure 2.1. The physical construction of a product element affects its characteristics and behaviors during use. For example, the physical construction of a product element influences the distribution of mechanical loads along the product element or the manner in which electricity or heat flows through it.

The product elements within a product must be designed so that they enable the product to satisfy all of its requirements, which includes performance, reliability, and cost. The entire set of requirements that must be satisfied comprises the *design requirements*, which are defined during product development. All elements of the design requirements are discussed later in this chapter.

The design requirements for a product are the basis for the design requirements of the product elements. Once the product elements' design requirements have been defined, it is possible to identify, evaluate, and select the materials that can be used for each product element.

There will be design trade-offs between the physical construction and the materials that can be used for a product element. Using a specific physical

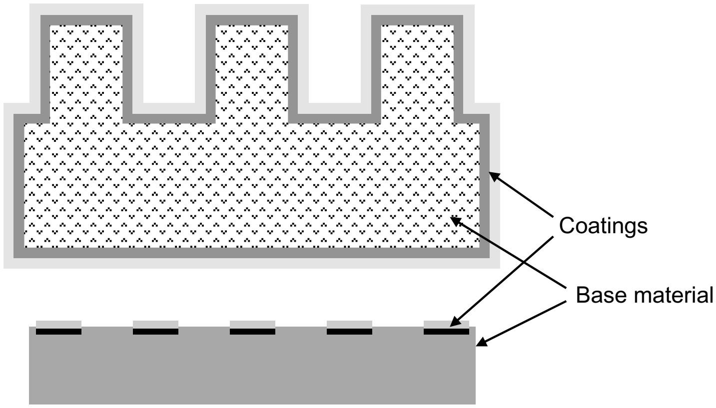


FIGURE 2.1

Schematics of components composed of a material with coatings applied over the surface.

construction can affect the range of materials that can be used for a product element. Alternatively, trying to use a specific material for a product element may constrain its physical construction. Ideally, a product element's physical construction and materials are optimized to provide the required performance and reliability at the lowest cost.

This chapter discusses how the design requirements for product elements are derived from the design requirements of the product. For the sake of the discussions that follow, we assume that a product that is being considered consists of more than one subassembly. Obviously, there are many products that have no subassemblies and that are composed of only product elements or that are a component. Examples are screws, electrical resistors, pencils, bottles, scissors, screwdrivers, and hammers. The discussions that follow can be extended to products like these.

2.2 DEVELOPING DESIGN REQUIREMENTS

The origin of a product's design requirements depends on whether the product is designed by a Type I or Type II company. The design requirements for a Type I product are based on the wants and needs of the intended customer. These wants and needs must be identified by the company making the product, and they are often communicated by the target customer in nontechnical and sometimes vague terms. For example, a product should not be too heavy, should be easy to open, or should look "high-tech." Design teams must convert these wants and needs to technical design requirements. For example, the mass of a product must be 0.5 to 0.7 kg, the force required to open a product should be less than 5 newtons, or a surface must reflect at least 80 percent of incident light.

Once the customer's wants and needs have been identified, the design team converts them to engineering requirements for the product. These engineering requirements become parts of the design requirements for the product. Successfully converting the customer's wants and needs to meaningful engineering requirements necessitates good communication between the marketing and engineering groups within a company. An engineering technique that is useful for helping design teams convert customer wants and needs to engineering requirements is quality function deployment, which is discussed in Chapter 7.

For a Type II product, the company ordering the product provides the Type II company with a set of design requirements. The Type II company designs its product based on these design requirements, which are already in technical terms. As discussed in Chapter 1, the client company, which can be either Type I or Type II, will use the item in its product.

After the design requirements for a product have been defined, the design team develops, evaluates, and selects product *design concepts*, which are descriptions of the product's physical form. After this the design team develops design concepts and defines design requirements for the subassemblies within the product. Subassemblies must be designed so that they satisfy the design requirements of the product. Finally, the design team develops design concepts and defines design requirements for the product elements within the subassemblies. The product elements must be designed so that they satisfy the design requirements of the subassemblies. A flowchart for this whole process is shown in Figure 2.2.

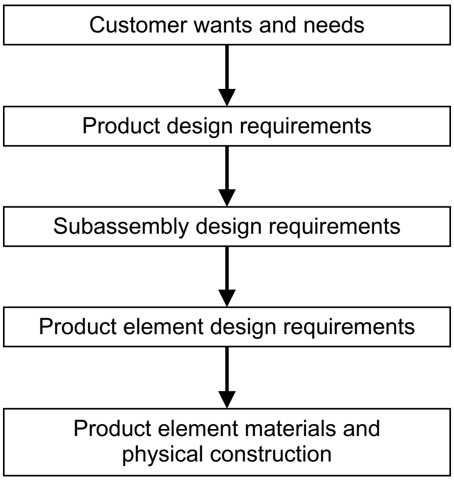


FIGURE 2.2

Flowchart for the design process.

Developing concept designs for a product, its subassemblies, and their product elements is an iterative process. Design teams evaluate the risks associated with the concept designs and their design requirements, and then they make modifications as needed to increase the likelihood of the product's success. This process is discussed in more detail in Chapters 7 through 10.

There can be more than one tier of subassemblies within a product—that is, subassemblies within subassemblies—so that there are sub-subassemblies, sub-sub-subassemblies, and so on. A schematic of this concept is shown in Figure 2.3. Figure 2.3(a) shows the highest-level subassemblies within the product, subassemblies 1, 2, and 3. Figure 2.3(b) shows two subassemblies within subassembly 3.

Each tier of subassemblies must be designed so that it satisfies the design requirements of the subassemblies in which it resides. For the product shown in Figure 2.3, subassemblies 3a and 3b are designed based on the design requirements for subassembly 3.

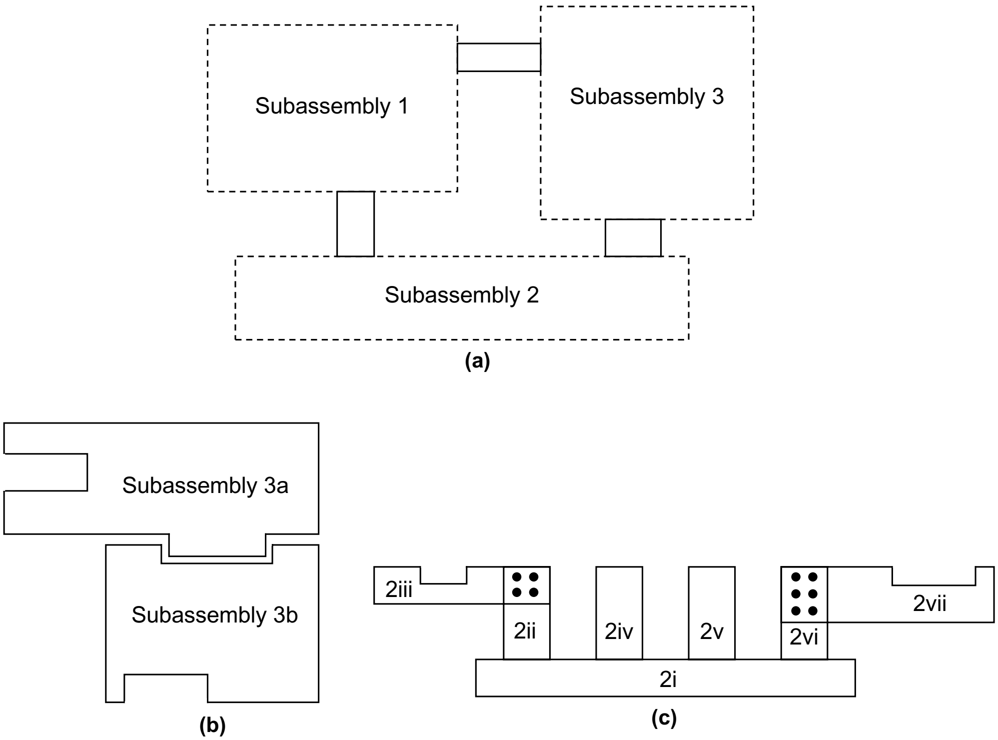


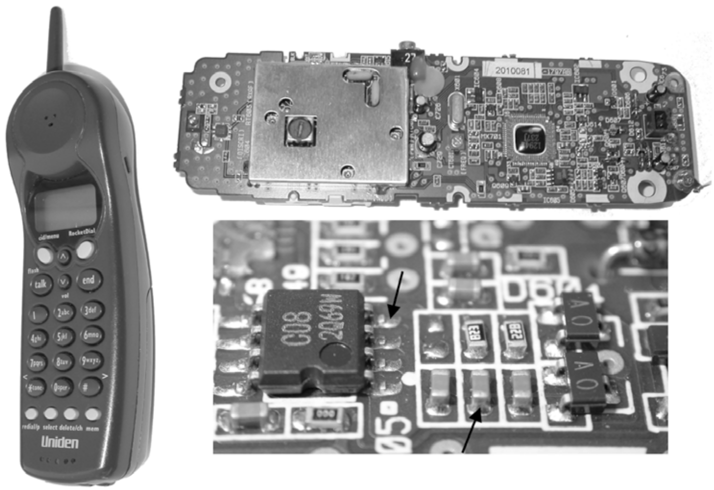
FIGURE 2.3

A product and its subassemblies. (a) Complete assembly, (b) subassembly 3 and its subassemblies, and (c) subassembly 2 and its product elements.

For ease of discussion, each subassembly, regardless of its level, will be referred to as a subassembly.

The following sections discuss the design requirements for products, subassemblies, and product elements. Five products will be used as examples. These products are an automobile, a cordless telephone, a motor, an industrial oven, and a cooking skillet. These products, and their subassemblies and product elements that will be used for the discussions, are shown in Figure 2.4.

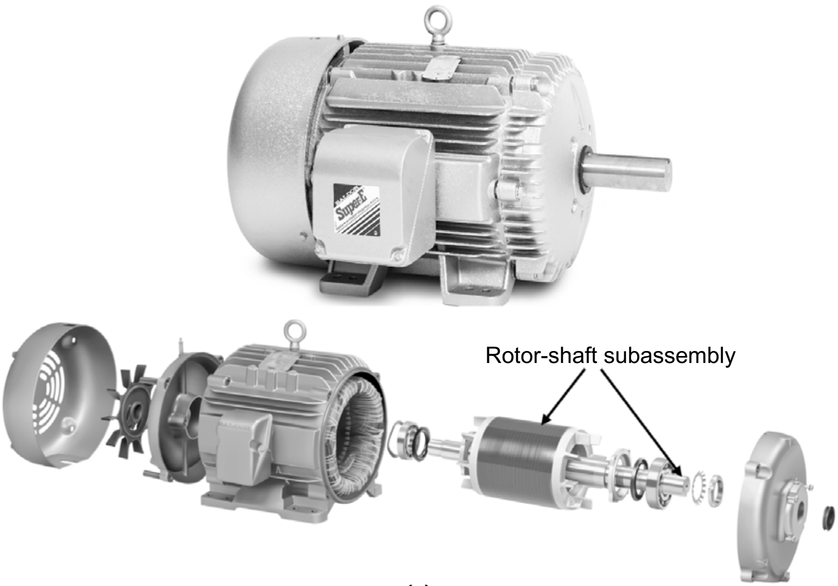
(a)



(b)

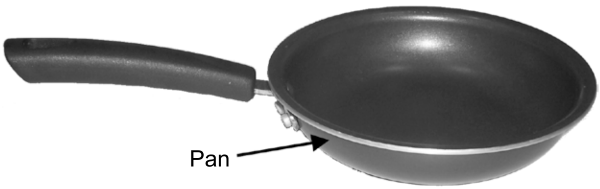
FIGURE 2.4

Example products, subassemblies, and product elements: (a) Automobile. (b) Cordless telephone. (*Continued*)



(c)

(d)



(e)

FIGURE 2.4 *Continued*

(c) Industrial oven. (d) Motor. (e) Skillet.

2.3 PRODUCT DESIGN REQUIREMENTS

Many products have groups of "customers" in addition to those that will be purchasing the product. These other customers are organizations that impose requirements, and although they will not be purchasing the product, it is necessary to satisfy their requirements. Such customers include the following:

- Manufacturing organizations that will be participating in the production of the product
- Industry organizations that set standards covering areas such as product performance, reliability, and safety
- Government organizations that regulate in a variety of areas such as product safety, manner in which products can operate, and the substances that can be used in products
- Legal organizations inside and outside of companies that oversee and control the use of intellectual property

It is important to be aware of all of the relevant nonpaying customers. Not doing so can lead to serious consequences, like a product being banned from sale or being recalled.

The general categories of product design requirements are as follows:

1. Performance requirements
2. Reliability requirements
3. Size, shape, mass, and style requirements
4. Cost requirements
5. Manufacturing requirements
6. Industry standards
7. Government regulations
8. Intellectual property requirements
9. Sustainability requirements

Each of these categories is discussed in more detail later in this section.

This section applies to Type I and Type II products that are assemblies or subassemblies. Type I and II products that are components will be addressed in the section on product element requirements. What follows is a description of each of the nine categories of product design requirements. More details about developing requirements for products are provided in Ulrich and Eppinger (2004).

2.3.1 Performance Requirements

The performance requirements describe the functions and features of a product. This involves assigning measurable target values for each performance attribute associated with a particular function or feature. It is important to identify attributes that are measurable; otherwise, it will be very difficult to objectively verify that the requirement is being met.

Some examples of the performance requirements are shown in Table 2.1 for the example products shown earlier in Figure 2.4. The XX's shown in the table take the place of the values that a design team would assign to each particular attribute. For an automobile, the design team would specify the time, measured in seconds, required for the automobile to accelerate from 0 to 100 kilometers per hour.

Table 2.1 Selected Performance Requirements of Example Products

Product	Performance Requirements
Automobile	▪ Acceleration: Time from 0 to XX kilometers/hour (seconds) ▪ Handling: Minimum turn radius at a specified speed (meters) ▪ Fuel economy: Driving distance per unit volume of gasoline (km/liter) ▪ Safety features: Air bag deployment time (seconds), braking distance at a specified speed (meters); visibility (glass light transmission, index of refraction) ▪ Comfort: Road and engine noise in passenger compartment (decibels)
Cordless telephone	▪ Transmission and reception power: XX milliwatts ▪ Screen readability: Size of characters (mm), brightness (lumens) ▪ Reception range: XX meters ▪ Battery: XX ampere-hours/cm^3
Motor	▪ Output power: XX watts ▪ Starting torque: XX newton-meters ▪ Full load torque: XX newton-meters ▪ Speed: XX revolutions per minute ▪ Noise level: XX decibels at a specific distance ▪ Control: On/off, electronic variable control
Industrial oven	▪ Exterior size: Length × width × height (cm) ▪ Capacity: Length × width × height (cm) ▪ Maximum operating temperature: XX°C ▪ Temperature uniformity: Temperature variation (°C) ▪ Temperature control: Set point or automated ▪ Heating efficiency: Energy required to heat up the chamber (watts/cm^3)
Cooking skillet	▪ Heating uniformity: Temperature variation (°C) ▪ Cooking surface properties: Nonstick or bare metal ▪ Maximum use temperature: XX°C ▪ Handle insulation: XX watt/m°C

The specific value of the acceleration depends on the wants and needs of the intended customer. Similarly, the target values for the automobile's remaining attributes (which go well beyond the list that is shown in Table 2.1) would be specified based on the wants and needs of the intended customer. The performance requirements for the mobile phone, motor, industrial oven, and skillet contain the same type of information for the attributes that are specific to each product.

The values assigned to each performance attribute correspond accurately to customer wants and needs. If they are not, then the product will end up being over- or underdesigned. In the former case, the product will have unnecessary extra costs because the product was designed to meet requirements that exceeded the customer's wants and needs. In the latter case, the product will not have the desired performance to satisfy the customer's wants and needs. Thus, it is worthwhile to invest the time and resources to properly convert the customer's wants and needs to engineering requirements.

Notice that none of the performance requirements for any of the products shown in Table 2.1 make any mention of the materials used in the products. However, as will be shown, a thorough consideration of all of the performance requirements of a Type I assembly or subassembly is crucial to properly select the materials that will be used to make the product elements.

2.3.2 Reliability Requirements

Reliability refers to a product's ability to perform as specified over a specific use period and under a given set of use conditions. The use period may be measured in different ways such as total ownership time, total on time, or total miles used. The use conditions consist of the mechanical and electromagnetic forces and the thermal, chemical, biological, electrochemical, and radiation environments to which a product is exposed during handling, shipping, and customer use. Examples of different use conditions are as follows:

- *Mechanical:* Static, dynamic, or cyclic loads; impact; rubbing
- *Electromagnetic:* Applied voltage, current, magnetic fields
- *Thermal:* Elevated temperatures; repeated cycling between temperature extremes
- *Chemical:* Gases (e.g., oxygen, nitrogen, or chlorine), liquids (e.g., solvents, acids, or bases)
- *Biological:* Body fluids
- *Electrochemical:* Metals near saltwater or acids
- *Radiation:* Ultraviolet light; radiation in nuclear reactors

The use conditions are associated with the (1) functionality of a product and (2) environment in which the product operates. The mechanical load on a motor as it moves another device and the electricity that passes through an electrical circuit are examples of use conditions associated with product functionality.

Examples of environmental exposure are the use of a motor in the presence of corrosive liquids and the use of an electrical circuit in a high-temperature environment. For products with subassemblies, the environmental conditions may be different from subassembly to subassembly. For example, in an automobile the maximum exposure temperature for a subassembly located in the engine compartment is much hotter than for a subassembly located in the passenger compartment.

Reliability is a concern because the materials that make up a product element can degrade as a result of exposure to the use conditions. The degradation causes changes in the properties of the materials, which can lead to a product element no longer meeting its design requirements, resulting in loss of performance of the product containing the product element.

When defining the use conditions, design teams may have to consider minor abuse and misuse.

Minor abuse refers to exposure to conditions that are somewhat excessive, the determination of which is subjective. Examples of minor abuse include dropping a telephone from 1-meter height onto concrete, driving an automobile over potholes, and small chemical spills onto a motor. Many products are expected to withstand a certain amount of minor abuse by the customer without failing to meet its performance requirements.

Misuse refers to exposure to conditions after which it is unrealistic to expect a product to function properly. For example, it is unrealistic for a customer to expect a cordless telephone to function properly after using it to hammer a nail into a wall or dropping it in a toilet.

A product that ceases to perform as designed is said to have failed. Failure to perform as specified involves the following two situations:

1. A particular function or feature degrades below its minimum performance requirements. The product may still function, but not as well as it did before the degradation occurred. An example is an oven taking too long to heat up or the buttons on a telephone requiring increased force for actuation.
2. A particular functionality is completely lost without warning. Sometimes, these failures are inconvenient, as for an electronic device that will not turn on or an automobile that will not start. In other cases, the loss of functionality is catastrophic, like when an airplane loses rudder control or a ladder collapses. Also, the failure may result in unexpected behavior of the product, resulting in harm to the user, such as when an electrical circuit shorts out and starts a fire.

Unexpected failures that occur during the normal, expected life of a product generally are due to the following five engineering problems:

1. *Inaccurate specification of the use conditions and reliability requirements for a product, its subassemblies, or product elements.* This occurs when a

design team does not thoroughly and accurately identify and quantify the performance and reliability requirements for a product, its subassemblies, or its product elements. Any of these can result in one or more product elements that are underdesigned for the application. The earlier in the design process that poorly defined requirements are made, the worse the problem will be. Underdefined product requirements will certainly result in underdesigned product elements, regardless of the skill of the engineers designing the product elements.

2. *Poor design.* This includes disregarding agreed-on performance and reliability requirements for a product, its subassemblies, or its product elements; not considering industry standards or government regulations; designing a product element with inadequate physical dimensions for withstanding exposure to the use conditions; and selecting suboptimum materials for a product element.
3. *Poor control of the variation of product elements' materials properties, shape, and size.* Even if the optimum materials are selected, too much variation of their properties and features can lead to product elements that do not have the required properties and therefore do not have the required reliability.
4. *Materials defects.* Defects in a material can compromise its properties to the point that it cannot withstand the exposure to the use conditions without substantial degradation. When this occurs, the affected product element fails to function as designed. Defects typically originate during manufacturing process and are discussed in Chapters 4 and 5.
5. *Product overstress.* Improper use of a product by a customer can cause stresses on a product element that exceed its design limits. Because overstress is a result of inappropriate use of the product, it is beyond the realm of design engineers. We will not be concerned with this item any further.

Ultimately, a product fails unexpectedly when there is an unexpected failure of one or more of its product elements.

Specifying the reliability requirements for a product involves identifying and, when possible, quantifying each use condition to which the product will be exposed. For example, a use condition for a cooking skillet might be exposure up to 250°C. Furthermore, design teams must identify the total exposure to each specific use condition that the product can withstand and still function with the specified performance. For the skillet, the total exposure to 250°C might be 5000 hours, which correlates to over 10 years of use if it is assumed that the skillet is used less than 2 hours a day.

Table 2.2 lists some of the product use conditions for the products discussed in the section on performance requirements. These conditions consider normal use and minor abuse, not misuse.

Table 2.2 Selected Products' Use Conditions

Product	Use Conditions
Automobile	■ Vibration: Up to XX mm displacement, XX m/s velocity, XX m/s^2 acceleration ■ Minimum and maximum temperatures: XX to YY°C ■ Humidity: Up to XX% relative humidity ■ Ultraviolet sunlight: XX mJ/cm^2 ■ Saltwater: Splashing from saltwater on road
Cordless telephone	■ Dropping: XX falls onto concrete from XX m height ■ Button pushing: XX times with YY N force ■ Maximum temperature: XX°C ■ Humidity: Up to XX% relative humidity ■ Contact with hand oils, food, and beverages
Motor	■ Torque: XX newton-meters applied by the motor load ■ Maximum temperature: XX°C ■ Vibration: Up to XX mm displacement, XX m/s velocity, XX m/s^2 acceleration ■ Moisture: Occasional splashing ■ Corrosive gases ■ Explosive gases and dust
Industrial oven	■ Minimum and maximum external temperature: XX to YY°C ■ Humidity: Up to XX% relative humidity ■ Mass of loads placed inside: Up to XX kg ■ Power surges: Up to XX volts ■ Corrosive gases
Cooking skillet	■ Maximum temperature: XX°C ■ Scraping with metal objects ■ Contact with detergents, food, hand oil ■ Dropping

2.3.3 Size, Shape, Mass, and Style Requirements

The size, shape, mass, and style requirements for a product are dictated by its functionality, ease of use, ability to fit into a particular space, and intended aesthetic appeal to customers. Style also includes shape, as well as color and surface texture. Table 2.3 lists some size, shape, mass, and style requirements for the example products.

2.3.4 Cost Requirements

The price at which a company thinks it can sell its product and the desired profit from each sale will set the requirements on the maximum allowed costs to design and manufacture the product. The cost to design a product includes the costs associated with the following:

Table 2.3 Selected Products' Size, Shape, Mass, and Style Requirements

Product	Size, Shape, Mass, and Style Requirements
Automobile	■ Size: XX cm × XX cm × XX cm (length × width × height) ■ Mass: XX grams ■ Shape: Sedan-like ■ Color: Metallic silver, black, or maroon
Cordless telephone	■ Size: XX cm × XX cm × XX cm (length × width × height) ■ Mass: XX grams ■ Color: Shiny black ■ Surface: Smooth
Motor	■ Size: XX cm × XX cm × XX cm (length × width × height) ■ Mass: XX grams ■ Color: Light blue
Industrial oven	■ Exterior Size: XX cm × XX cm × XX cm (length × width × height) ■ Capacity: XX cm × XX cm × XX cm (length × width × height) ■ Surface: Brushed metal
Cooking skillet	■ Mass: XX grams ■ Diameter of pan: XX centimeters ■ Height of pan: XX centimeters ■ Handle length: XX centimeters ■ Handle shape: Tapered ■ Handle color: Black

- Engineering personnel
- Design tools such as computer-aided design and finite element analysis
- Making product prototypes for testing and evaluation
- Product testing
- Materials characterization and materials reliability testing

The cost to manufacture a product includes the costs for the following:

- Purchasing materials, components, and subassemblies used to build the product
- Manufacturing components and subassemblies used within the product
- Assembling the product from the components or subassemblies
- Testing to verify that a product meets the design requirements
- Packaging and shipping the product to customers and stores
- Providing warranty and other product support to customers

The total allowable costs to manufacture a product will constrain the allowable costs for the materials, components, subassemblies, and manufacturing processes used to make the product. None of these take into consideration the costs associated with problems such as failed product tests during product development and poor manufacturing yields.

2.3.5 Industry Standards

Industry standards are documents that contain requirements that have been agreed on by groups of companies and people working in specific industries or on specific types of products. The standards address product performance, safety, reliability, and the methods for evaluating product performance, reliability, and safety.

There are organizations within particular industries and across industries that oversee the development and revision of standards. Some of the organizations that write standards and the products to which the standards apply are shown in Table 2.4. This is a fraction of the standards-writing organizations that exist worldwide. Some of the organizations listed may seem obscure. However, for most of these organizations, the total sales of the products that they oversee are substantial. All of the organizations shown have websites, and most of them have search engines for finding relevant standards. A larger list of standards-writing organizations is given in the *ASM Handbook* (1997). Some of the applicable standards for the example products used in this chapter are shown in Table 2.5.

2.3.6 Government Regulations

Government regulations place requirements on things such as the manner in which a product operates, product safety, and substances that can and cannot be used in the product. Table 2.6 lists some of the agencies in the United States and Europe that write regulations that apply to products. Other countries have their own agencies that write regulations.

Design teams must be knowledgeable of and understand the applications of the government regulations that apply to their products. Failure to comply with existing regulations can result in a product being banned from sale. Table 2.7 lists some of the government regulations that apply to the example products.

2.3.7 Intellectual Property

Intellectual property is a product of an individual's intellect and includes patents, trade secrets, copyrights, and trademarks. A patent is a right given by a government to an inventor (or inventors). The government gives the inventor the sole right to make, use, or sell a product covered by the claims of an issued patent to the inventor(s). The invention must be a process, machine, manufacture, or composition of matter. For a patent to be issued, the invention must meet the government's definition of novel, useful, and nonobvious.

The aesthetic design of a product can also be protected by a design patent, preventing others from using a similar product's nonfunctional styling. If the inventor works for a company, the inventor has usually assigned the rights to the

Table 2.4 Selected Industry Standards-Writing Organizations and Applicable Products or Industries

Organization	Applicable Product or Industry
American Boat and Yacht Council	Safety standards for boat building and repair
American Gear Manufacturers Association	Gears
American National Standards Institute	Covers a wide range of products in different business sectors
ASME International	Boiler and pressure vessels Nuclear components
Association for the Advancement of Medical Instrumentation	Medical devices
ASTM International	Specification and testing of a wide range of materials for a wide range of applications.
Institute of Electrical and Electronics Engineers	Electronics and electrical devices
International Electrotechnical Commission	Electrical, electronic, and related technologies
International Organization for Standardization	Covers a wide range of products and applications
IPC	Electronic circuit board manufacturing and assembly
Metal Powder Industries Federation	Powder metal components
National Electrical Manufacturers Association	Electrical equipment
NSF International	Food, water, and consumer products
SAE International	Aerospace and ground vehicles (e.g., automotive, agricultural, off-road machinery, construction), trucks and buses
The JEDEC Solid State Technology Association	Semiconductor components
Underwriters Laboratories	Fire and explosion safety of electrical products

patent to his or her employer as a condition of employment. The different types of intellectual property are as follows.

A *patent* right is a negative right in that the patent holder can prevent others from practicing any of the claims in the invention; however, it does not give the patent holder the right to practice the invention described in the claims because doing so could infringe on someone else's patent rights or be against

Table 2.5 Selected Industry Standards Applicable to Example Products

Assembly/ Subassembly	Applicable Industry Standards
Automobile	▪ SAE J299 Stopping Distance Test Procedure ▪ SAE J144 Subjective Rating Scale for Vehicle Handling ▪ SAE J850 Fixed Rigid Barrier Collision Tests
Cordless telephone	▪ TIA-1083 Telephone Terminal Equipment Handset Magnetic Measurement Procedures and Performance Requirements ▪ TIA/EIA-470-C.310 Cordless Telephone Range Measurement Procedures
Motor	▪ IEC 60034 Rotating Electrical Machines ▪ IEEE 114 Standard Test Procedure for Single-Phase Induction Motors ▪ NEMA MG1 Motors and Generators
Industrial oven	▪ National Fire Protection Association (NFPA) 86: Standard for Ovens and Furnaces ▪ NEMA 250 Enclosures for Electrical Equipment
Cooking skillet	▪ NSF/ANSI 51 Food Equipment Materials

Table 2.6 Selected Government Regulatory Agencies

Government Agency	Applications
United States Federal Communications Commission	Products that emit electromagnetic radiation
United States Federal Aviation Administration	Aviation products
United States Nuclear Regulatory Commission	Products that use radioactive materials
United States Occupational Safety & Health Administration	Product safety
United States Consumer Products Safety Commission	Consumer safety
United States Food and Drug Administration	Medical devices
European Union	Many

government policy. An inventor can license or sell the right to use the patent to others.

A *trade secret* is information that (1) derives independent economic value from not being generally known or readily ascertainable to other persons who can obtain economic value from its disclosure or use and (2) is the subject of efforts that are reasonable under the circumstances to maintain its secrecy. Examples

Table 2.7 Selected Products and Applicable Government Regulations

Assembly/Subassembly	Regulation
Automobile	▪ Directive 2000/53/EC (European Parliament) End-of life vehicles: Regulates the substances restricted for use in automobiles ▪ U.S. Code of Federal Regulation Title 49 Part 571 Federal Motor Vehicle Safety Standards: Regulates motor vehicle safety requirements ▪ U.S. Clean Air Act: Regulates the types and amounts of emissions from automobile
Cordless telephone	▪ U.S. Code of Federal Regulation Title 47 Telecommunication Part 15 Radio Frequency Devices Council: Regulates the allowable electromagnetic radiation emitted from electronics ▪ Directive 2002/95/EC (European Parliament) Restriction of the use of certain hazardous substances in electrical and electronic equipment: Regulates the substances restricted for use in electronic devices
DC motor	▪ U.S. Code of Federal Regulation Title 10 Energy Part 431 Energy Efficiency Program for Certain Commercial and Industrial Equipment: Regulates energy conservation requirements for electric motors ▪ U.S. Code of Federal Regulation Title 47 Telecommunication Part 15 Radio Frequency Devices Council: Regulates the allowable electromagnetic radiation emitted from electronics
Industrial oven	▪ U.S. Code of Federal Regulation Title 47 Telecommunication Part 15 Radio Frequency Devices Council: Regulates the allowable electromagnetic radiation emitted from electronics
Cooking skillet	▪ U.S. Code of Federal Regulation Title 21 (Federal Food, Drug and Cosmetic Act regulations): Regulates the use of nonstick coatings ▪ Regulation (EC) No. 1935/2004 of the European Parliament and of the Council of 27 October 2004 on materials and articles intended to come into contact with food: Regulates the use of cooking utensil materials

of trade secrets can be formulas for products, methods used in manufacturing, and compilations of information.

A *copyright* protects the embodiment of a work (e.g., a work of art, software, or a performance), not the inventive ideas behind the work.

A *trademark* or *service mark* is a word, phrase, symbol, design, or combination of these that identifies and distinguishes the source of the goods or services of one party from those of others.

One effect of patents owned by other companies is to restrict the implementation of certain functions within a product or restrict the means for achieving the functions and features. By the way, it is possible for a patent for an unrelated type of product to apply to the functions or features that are intended within the product being designed. For the case of a patent covering the implementation of certain functions, the design team must decide to either purchase the right to use the invention covered by the patent or exclude the functionality from the product. For the case of a patent covering the means for achieving certain functions or features, the design team must come up with a new invention for achieving the functions or features, purchase the rights to use the invention covered by the patent, or exclude the functions and features from the product.

It is best to gain the knowledge about patent coverage at the beginning of a design effort. Design teams can conduct patent searches before starting the design of a new product. Resources for researching patents are available on the Internet. Some of the various worldwide patent offices and their websites for researching patents are the following:

- U.S. Patent and Trade Office, *www.uspto.gov*
- European Patent Office, *www.european-patent-office.org*
- Japan Patent Office, *www.jpo.go.jp*

2.3.8 Manufacturing Requirements

Companies that have internal manufacturing capabilities may require that specific manufacturing processes and materials be used to produce the product. For example, these constraints can restrict the product's shape and dimensions so that the existing manufacturing equipment can handle the product as it is being built. Perhaps the manufacturing capabilities may restrict the types of joints that can be used to join components and subassemblies, which will have an impact on the materials that can be used to form the joints.

For a product that is similar to past products, manufacturing constraints such as these are acceptable. In fact, they may be desirable because they provide a certain level of confidence in the design of products that can be produced using familiar processes and materials.

In cases where a new product differs significantly from past products, the constraints of using specific materials and processes may provide a competitive advantage or be a burden. It can be a competitive advantage if the design team is creative enough to develop a product around the manufacturing requirements. However, the requirements are a burden if the existing processes are inherently incompatible with the new product or if the design team does not have the required creativity.

2.3.9 Sustainability Requirements

Sustainability means "meeting the needs of the present without compromising the ability of future generations to meet their own needs." It requires that human activity only utilizes nature's resources at a rate at which they can be replenished naturally. The needed aim of sustainable design is to manufacture products in a way that reduces use of nonrenewable resources, minimizes environmental impact, and relates people with the natural environments.

Sustainable design (also referred to as “green design,” “eco-design,” or “design for environment”) is the art of designing physical objects to comply with the principles of ecological sustainability. Sustainable technologies are those that use less energy, use fewer limited resources, do not deplete natural resources, do not directly or indirectly pollute the environment, and can be reused or recycled at the end of their useful life.

Sustainable design is a general reaction to the rapid growth of economic activity and human population, depletion of natural resources, damage to ecosystems, and loss of biodiversity. It is considered a means of reducing the use of nonrenewable resources and our impact on the environment, while maintaining quality of life, by using clever designs to substitute less harmful products and processes for conventional ones.

The following are some common principles of sustainable design:

- Use low-impact materials. Choose nontoxic, sustainably produced, or recycled materials that require little energy to process.
- Use manufacturing processes that require less energy.
- Make longer-lasting and better-functioning products that will have to be replaced less frequently, reducing the impact of producing replacements.
- Design products for reuse and recycling.
- Use materials that come from nearby (local or bioregional), sustainably managed renewable sources that can be composted when their usefulness has been exhausted.

For example, automobiles and appliances can be designed for repair and disassembly (for recycling) and constructed from recyclable materials such as steel, aluminum, and glass, and from renewable materials (e.g., wood and plastics from natural feedstock).

From the standpoint of product design requirements, sustainability requirements are currently based on whether or not it is important to customers that their purchase of the product is environmentally responsible. Thus, it is actually an aspect of the customers’ wants and needs that influences the materials and processes that can be used. This in turn will affect the design of the product and its subassemblies.

2.4 SUBASSEMBLY DESIGN REQUIREMENTS

The process for identifying the design requirements for a product’s subassemblies is similar to that described in the previous section for the product. The same design requirements discussed there must be defined for each subassembly in a product. The performance and reliability requirements for one subassembly per product for the example products used earlier are shown in Tables 2.8 and 2.9.

Table 2.8 Selected Subassemblies and Their Performance Requirements

Subassembly	Performance Requirements
Automobile windshield wiper blade	■ Push water off windshield: XX% visibility ■ Support loads from wiper arm: XX kg force
Cordless telephone circuit board	■ Process transmission and reception signals: XX instructions per second ■ Information loss: < XX processing errors ■ Battery power consumption: XX ampere-hours ■ Command processing: redial, volume control, and so on ■ Memory: Store XX phone numbers
Motor rotor-shaft	■ Carry current: XX amperes ■ Transmit torque to load: XX newton-meters
Industrial oven heating element	■ Heat oven up: > XX°C ■ Produce uniform heating: < XX°C variation
Cooking skillet	■ No subassemblies

Table 2.9 Selected Subassemblies and Their Use Conditions

Subassembly	Use Conditions
Automobile windshield wiper blade	■ Maximum temperature: +YY°C ■ Up to XX% relative humidity ■ Ultraviolet sunlight ■ Splashing from saltwater on road ■ Windshield wiper fluid
Cordless telephone circuit board	■ Impact: Drop 2 meters onto concrete ■ Maximum temperature: +YY°C ■ 85% relative humidity ■ Contact with food, beverages, hand oils, lotions, and other household chemicals
Motor rotor-shaft	■ Maximum temperature: +YY°C ■ Cyclic loading: Torsion loads between –XX and +XX N-m ■ Vibration: Up to XX mm displacement, XX ns velocity, XX m/s acceleration ■ Moisture, corrosive liquids, corrosive gases ■ Explosive gases and dust ■ Dust ■ Ignitable fibers
Industrial oven heating element	■ 1500°C ■ Thermal cycling: Between 25°C and 1500°C ■ Corrosive vapors
Cooking skillet pan	■ No subassembly

2.5 PRODUCT ELEMENT DESIGN REQUIREMENTS

The general categories for product element design requirements are the same as those for assemblies and subassemblies. This section discusses each of these categories as they apply to product elements, using the following product elements as examples:

- Automobile windshield wiper blade insert
- Cordless telephone circuit board solder joint
- Motor shaft
- Industrial oven heating element
- Cooking skillet pan

2.5.1 Performance Requirements

Each product element within an assembly or subassembly has a specific set of functions that enable the functionality and performance of an assembly or subassembly. The performance requirements describe the attributes that the product element must have to function as required. The attributes can be described in terms of mechanical, electromagnetic, thermal, optical, physical, chemical, electrochemical, and cosmetic properties.

The performance requirements for the product elements within the example products and subassemblies discussed earlier are shown in Table 2.10. The product elements were shown earlier in Figure 2.4. Each performance require-

Table 2.10 Selected Product Element Performance Requirements

Product Element	Performance Requirements
Automobile wiper blade insert	▪ Conform to the surface of the windshield: Elastic modulus < XX MPa ▪ Not deform too much: Elastic modulus > YY MPa ▪ Slide easily across the surface of the windshield with low noise: coefficient of friction < X, unitless
Cordless telephone circuit board solder joint	▪ Conduct electricity between circuit board and electrical component: < XX ohms ▪ Hold component in place on the circuit board: Strength > XX MPa
Motor shaft	▪ Transfer power from rotor to load: Strength > XX MPa and elastic modulus (rigidity) > XX MPa
Industrial furnace heating element wire	▪ Heat oven to maximum temperature: Electrical resistance XX–YY ohms)
Cooking skillet pan	▪ Hold food: Strength > XX MPa and elastic modulus (rigidity) > XX MPa ▪ Uniform heating: Thermal conductivity > XX watts/m · K ▪ Nonstick surface: Surface energy < XX ergs

ment consists of an attribute and the required value for the attribute. The lower and upper bounds of the required property are indicated by X and Y. The value is stated as a minimum, maximum, or range. For example, the wiper blade insert must be flexible enough so that it conforms to the surface of a windshield, but not so flexible that it deforms too much when pressed against the windshield. Also, the insert must slide easily across the surface of a windshield. The flexibility requirements can be described by the elastic modulus of the insert material, which can have a range of values that correspond to acceptable flexibility. The sliding requirements can be described by the coefficient of friction of the insert material, which must be below a maximum value. The same approach to specification of the product element performance requirements is applied to the telephone solder joint, motor shaft, industrial furnace heating element wire, and cooking skillet pan.

There is one significant difference between the design requirements for products and subassemblies and the design requirements for product elements. The performance requirements for products and subassemblies are described only in terms of the functionality that they provide, whereas the performance requirements for product elements are described in terms of material properties (e.g., strength, electrical conductivity, melting-point temperature). This is significant because these material properties, along with the other product element design requirements, will be used as part of the criteria for identifying and selecting materials that can be used to form a product element.

2.5.2 Reliability

The reliability of a product element refers to its ability to function as required over a specific use period when exposed to a specific set of use conditions. The reliability of a product element depends on its materials, the use conditions to which it is exposed, and the response of the materials to the use conditions. The use conditions can cause degradation of the materials to the point that the product element no longer performs as required. The degradation can occur quickly or over a long period of time. In either case, the product element fails once the material degrades to the point at which the product element no longer performs as required. The manner by which materials degrade and the effects of specific use conditions on materials is discussed in Chapter 6.

Within an assembly or subassembly there will be different use conditions from product element to product element. The specific use conditions for a product element are related to the following three items:

1. The *functionality of a product element*. While performing its function, a product element is exposed to mechanical, electromagnetic, thermal, chemical, biological, electrochemical, or radiation conditions. The mechanical forces acting on the tip of a wiper blade are due to the force of the wiper subassembly pushing the blade against the windshield and the motion of the blade across the windshield. The on and off cycling of an industrial oven will expose the

heater element wire to thermal stress fatigue. Also, the high temperatures will expose the heater element wire to oxidizing conditions.

2. The *physical location of a product element within an assembly or subassembly.* The product elements within a product can be exposed to different environmental use conditions depending on their physical location within the product. A product element located within the passenger compartment of an automobile is not exposed to the same temperature and corrosion conditions as a product element located in the engine compartment. The pan of a skillet is exposed to much higher temperatures compared to the handle. The differences in the specific environment require a design team to carefully consider the use conditions for every product element in a product.
3. *Interactions between product elements.* There can be interactions between product elements that are not directly related to how they function. If the materials are incompatible, then the interactions can lead to degradation of one or both of the product element's materials. For example, galvanic corrosion can result when two metallic product elements are in contact in the presence of moisture. Another example is a product element made of a polymer or rubber material that, when exposed to elevated temperatures, releases a gaseous chemical that reacts with other product elements located nearby. When selecting materials, design teams must understand the responses of the different materials to the environmental conditions and the effects of the responses on the other materials in the vicinity.

Example product elements' use conditions are shown in Table 2.11.

2.5.3 Size, Shape, and Mass Requirements

The size, shape, and mass requirements for a product element will have a huge influence on the materials that can be used. Consider a component that must carry five amperes of current without heating up by more than 15°C above the ambient temperature. The electrical conductivity for a component with a 1-mm diameter must be about four times greater than the electrical conductivity for a component that can be 2 mm in diameter. A bicycle frame that has to be 8 kg must have frame tubes made of a lower-density material compared to a 12-kg frame. For a component that needs to support 100 kg, the yield stress for the material in a component that must be 5 mm in diameter has to be much greater compared to the material in a component that can be 10 mm in diameter.

The product element size, shape, and mass requirements for the example product elements are shown in Table 2.12.

2.5.4 Cost

The cost to form a product element or purchase a component depends on the factors that follow.

Table 2.11 Selected Product Elements and Their Use Conditions

Product Element	Use Conditions
Automobile wiper blade insert	■ Pressure against windshield: XX newtons ■ Cyclic motion: Back and forth ■ Temperature extremes: XX–YY°C ■ Relative humidity: XX% ■ Ultraviolet radiation, wiper fluid, saltwater
Cordless telephone circuit board solder joint	■ Temperature extremes: XX–YY°C ■ Relative humidity: XX%
Motor shaft	■ Torsion loads: XX newton-meters ■ Bending loads: XX newtons ■ Temperature extremes: XX–YY°C ■ Corrosive liquids and gases
Industrial furnace heating element wire	■ Temperature extremes: XX–YY°C ■ Thermal cycling ■ Chemical vapors
Cooking skillet pan	■ Temperature extremes: XX to YY°C ■ Scraping surface ■ Dropping 1 meter onto tile floor ■ Food, detergents, hot water

Table 2.12 Selected Product Element Size, Shape, and Mass Requirements

Product Element	Size, Shape, and Mass Requirements
Automobile wiper blade insert	■ Length: XX mm ■ Shape: Cross-section shape (e.g., T-shaped)
Cordless telephone circuit board solder joint	■ Shape: Fillet
Motor shaft	■ Length: XX centimeters ■ Diameter: XX centimeters
Industrial furnace heating element wire	■ Length: XX centimeters ■ Diameter: XX centimeters
Cooking skillet pan	■ Diameter: XX centimeters ■ Mass: XX grams

The materials that constitute a product element. As the design requirements for a product element become more restrictive, the range of materials options available to the design team decreases and the cost of the materials that can satisfy the requirements increases. Looking at just performance and reliability, materials that result in low product element performance and reliability will be the least expensive compared to materials that enable greater product element performance and reliability.

The manufacturing processes used to form a product element. Some processes are more expensive than others because of the process complexity, equipment required, skill required to operate the processes, or the number of people required to perform the work. There is not a direct relationship between the complexity and cost of a manufacturing process and the performance and reliability requirements of a component. There are many low-complexity processes used to fabricate high-performance and high-reliability components.

Whether a component is custom made or purchased from an "off-the-shelf supplier." Off-the-shelf components typically cost less than custom-designed components because they are made in higher quantities compared to custom ones. The higher volumes enable off-the-shelf components to be manufacturered at lower prices compared to custom-made components.

The quantity of components or materials being purchased. Typically, as the number of components being purchased increases, the cost per piece decreases. This is also true for materials used to form components, joints, and in-process structures.

Quality problems associated with a component or material. Many companies seek out the lowest cost supplier of components and materials, and sometimes neglect consideration of the potential costs associated with manufacturing quality and product quality problems. The technical capabilities and skill of the supplier of a component or material will have an impact on quality. If the performance and reliability of a supplier's product are poor or inconsistent, then there will be additional, unplanned costs associated with poor manufacturing quality, customer returns, warranty repairs, and product recalls.

2.5.5 Industry Standards

Just as there are industry standards for assemblies and subassemblies, there are also industry standards that address the performance and reliability of product elements. In some cases, a specific standard will discuss assembly, subassembly, and product element requirements. Standards specific to product elements address issues such as the following:

- The size and shape of product elements used for specific applications
- The materials that can and cannot be used for product elements used for specific applications
- The tests required to verify the properties of the materials used to make a product element

The standards-writing organizations discussed earlier also apply to standards for product elements. Table 2.13 lists selected product elements and examples of applicable standards.

Table 2.13 Product Elements and Selected Applicable Industry Standards

Product Element	Applicable Industry Standards
Automobile wiper blade insert	▪ None found
Cordless telephone circuit board solder joint	▪ IPC 9701A Performance Test Methods and Qualification Requirements for Surface Mount Solder Attachments ▪ IPC J-001D Requirements for Soldered Electrical and Electronic Assemblies ▪ IPC-A-610D Acceptability of Electronic Assemblies
Motor shaft	▪ None found
Industrial furnace heating element wire	▪ ASTM F289 Standard Specification for Molybdenum Wire and Rod for Electronic Applications ▪ ASTM B76 Standard Test Method for Accelerated Life of Nickel-Chromium and Nickel-Chromium-Iron Alloys for Electrical Heating
Cooking skillet pan	▪ NSF/ANSI 51 Food Equipment Materials

2.5.6 Government Regulations

Government regulations regarding components used in assemblies and subassemblies are typically related to requirements on the materials from which components can and cannot be made. The requirements address concerns about the toxicity of the substances used in a product element, the reliability of the materials in specific applications, and the impact of the substances used on the environment after the product has been thrown away. Every country has its own set of regulations.

One example is European Union legislation (Directives 2000/53/EC, 2002/95/EC, 2002/96/EC) that restricts the use of certain substances from certain types of electrical and electronics products. Some of the restricted substances are lead, mercury, and cadmium. For the products covered by the legislation, the amounts of these substances must be below certain levels in the materials that make up individual product elements. Thus, these regulations affect the materials' options available to make individual product elements.

2.5.7 Intellectual Property

There are many patents regarding the design and manufacture of product elements. Just as for an assembly or subassembly, if a patent is found that is applicable to the product element being selected or designed, then the design team has to decide whether to license the patent or engineer the product element to avoid conflict with the patent.

2.5.8 Manufacturing Requirements

Type I and Type II companies may require that specific processes be used for fabricating components and building assemblies or subassemblies. One reason for the restrictions is that a company has internal manufacturing capabilities that must be used, and there is no desire to invest in additional manufacturing capabilities. Another reason is that a company has experience with and is comfortable with product elements fabricated using a familiar manufacturing process, and wants to reduce the risk for problems when developing a new product.

Restrictions on the processes that can be used will restrict the materials that can be used to make product elements because the materials must be compatible with the processes. For example, components that must be joined using a specific welding, brazing, or soldering process must be made out of materials that enable good joints to be formed using the joining process. This may exclude the use of off-the-shelf components from one or more suppliers because the components are made of materials that are incompatible with the process. For a custom component, the restriction may require the use of certain materials in order to form a good joint.

Restricting the manufacturing process to only familiar ones will restrict the options of materials that can be used to form a product element because in many cases a specific set of materials can be used with a particular process. For example, only certain aluminum, zinc, or magnesium alloys can be used with die casting. In some respects, these constraints are acceptable, and may in fact be desirable, because they provide confidence in the design of components, subassemblies, and products produced using processes and materials with which there is experience. In cases when a new product is significantly different than older products, the constraints of using specific manufacturing processes may seem to be a burden.

2.5.9 Sustainability Requirements

The sustainability requirements for the product become the sustainability requirements for the product elements. These requirements restrict the materials that can be used in product elements to materials such as those that can be reused or recycled. The requirements will also restrict the manufacturing processes that can be used to form product elements to processes that do not harm the environment and do not use chemicals and materials manufactured using environmentally unfriendly processes.

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